「Newton Iteration.

「Prop. Let $\alpha > 0$ be fixed. Given $x_1 > \sqrt{\alpha}$, define recursively

$$x_{n+1} = \frac{1}{2}\cdot\left(x_n + \frac{\alpha}{x_n}\right) \quad (n \in \mathbb{N})$$

Then, $\{x_n\}$ is monotonically decreasing and converges to $\sqrt{\alpha}$.

Proof. Observe that: for each $n \in \mathbb{N}$,

$$\frac{1}{2}\cdot\left(x_n + \frac{\alpha}{x_n}\right) = x_{n+1} \leq x_n \iff \alpha \leq x_n^2 \quad \text{if } x_n > 0.$$

Clearly, all $x_n > 0$, by definition, so using $\alpha \leq x_n^2$ for all $n \in \mathbb{N}$, by induction.

For $n=1$, $\alpha \leq x_1^2$, since $\sqrt{\alpha} < x_1$. So, $x_2 \leq x_1$.

If $\alpha \leq x_n^2$, then

$$x_{n+1} = \frac{1}{2}\cdot\left(x_n + \frac{\alpha}{x_n}\right) \leq \frac{1}{2}\cdot(x_n + x_n) = x_n$$

and,
$$\frac{1}{2}\left(x_n + \frac{\alpha}{x_n}\right) \geq \sqrt{x_n \cdot \frac{\alpha}{x_n}} = \sqrt{\alpha}, \quad \text{So } \sqrt{\alpha} \leq x_{n+1} \leq x_n.$$

So that, the equivalent condition with the induction gives that

$$x_{n+1} \leq x_n \quad \text{for all } n \in \mathbb{N}$$

That is, decrease monotonically, and $x_n \geq \sqrt{\alpha}$ for all $n \in \mathbb{N}$

Now, since $\{x_n\}$ is bounded below by $\sqrt{\alpha}$ and decreasing, $\{x_n\}$ has a limit, denote $L = \lim_{n\to\infty} x_n$. Then,

$$L = \lim_{n\to\infty} x_{n+1} = \lim_{n\to\infty} \frac{1}{2}\left(x_n + \frac{\alpha}{x_n}\right) = \frac{1}{2}L + \frac{\alpha}{2L} \Rightarrow L = \sqrt{\alpha}.$$

So, proved ∎

Note: Moreover, we can measure the 'error': Put $\varepsilon_n = |x_n - \sqrt{\alpha}| = x_n - \sqrt{\alpha}$.

Then, for each $n \in \mathbb{N}$, $\varepsilon_{n+1} = \dfrac{\varepsilon_n^2}{2x_n} < \dfrac{\varepsilon_n^2}{2\sqrt{\alpha}}$ and that $\varepsilon_{n+1} < 2\cdot\sqrt{\alpha}\cdot\left(\dfrac{\varepsilon_1}{2\sqrt{\alpha}}\right)^{2^n}$.

Since
$$\varepsilon_{n+1} = x_{n+1} - \sqrt{\alpha} = \frac{1}{2}\cdot\left(x_n + \frac{\alpha}{x_n}\right) - \sqrt{\alpha} = \frac{1}{2}\cdot\left(\varepsilon_n + \sqrt{\alpha} + \frac{\alpha}{\varepsilon_n + \sqrt{\alpha}}\right) - \sqrt{\alpha}$$
$$= \frac{1}{2}\cdot\left(\frac{\varepsilon_n^2}{\varepsilon_n + \sqrt{\alpha}}\right) = \frac{\varepsilon_n^2}{2x_n} \leq \frac{\varepsilon_n^2}{2\sqrt{\alpha}}, \quad \text{so that}$$

$$\varepsilon_n \leq \frac{1}{2\sqrt{\alpha}}\cdot\varepsilon_{n-1}^2 \leq \frac{1}{2\sqrt{\alpha}}\cdot\left(\frac{1}{2\sqrt{\alpha}}\right)^2\cdot\varepsilon_{n-2}^{2^2} \leq \cdots \leq \left(\frac{1}{2\sqrt{\alpha}}\right)^{2^n - 1}\cdot\varepsilon_1^{2^{n-1}}$$

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